



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Summer Term: 2026-27

Course Handout Part II

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number: FIN F414

Course Title: Financial Risk Analytics and Management

Instructor-in-charge: Prof. Byomakesh Debata

Team of Instructors: Prof. Byomakesh Debata

1. Course description:

This course focuses on the concept of financial risk in modern finance. Despite market volatility and irrational exuberance, effective risk quantification and management remain crucial. Regulators have imposed safeguards to curb excessive risk-taking by financial institutions. As a result, risk management has become a core function in the finance industry. The course is designed to prepare students for the growing demand in financial risk management roles.

Financial risk (which includes interest rate risk, credit risk, foreign exchange risk and portfolio risk), Application of Asset price Theory, CAPM and Modern Portfolio Theory, Multifactor models of risk and return, Capital adequacy, BASEL Norms; Comprehensive Capital Analysis and Review, Basel Committee on Banking Supervision guidelines, Value at Risk (VaR), VaR from regulator perspective, Responsive VaR Model, Addressing VaR weakness, Back testing Concepts - Checking the Strength of VaR Model, Issues with the Standard VaR models, Risk Not in VaR (RNIV) to VaR migration of risk, Volatility Clustering, Quantifying and Modeling volatilities, Hedging techniques, Options Greeks, Managing Risk with Derivatives.

2. Scope and Objective:

The three core focus areas dealt with in this course are (i) Risk Identification, (ii) Risk Measurement, and (iii) Risk Management, in the context of modern-day finance. The scope and objective of the course is to enable students learn essentials of financial risk management. The three core focus areas along with the functions and workings of financial institutions encompass the entire coverage of this course. To meet the stated objective of the course, a detailed course plan is designed which students can refer in section 6 of this document.

3. Course learning outcomes:

- a. Understand and classify financial risks such as interest rate risk, credit risk, foreign exchange risk, and portfolio risk in modern financial markets.
- b. Analyze and implement risk measurement tools such as Value at Risk (VaR), Responsive VaR, and backtesting techniques.
- c. Interpret and evaluate regulatory frameworks like Basel norms and its guidelines for capital adequacy and systemic risk control.
- d. Develop risk mitigation strategies using derivatives, including hedging techniques and options Greeks, to manage exposure effectively.



4. Text books and references:

- TB1.** Hull, J. C., *Risk Management and Financial Institutions*, 4th Ed., Wiley
RB1. Jorion, P. (2007). *Value at risk: The new benchmark for managing financial risk* (3rd ed.). McGraw-Hill.
RB2. Hull, J. C. (2015). *Options, futures, and other derivatives* (9th ed.). Pearson Education.
RB3. Crouhy, M., Galai, D., & Mark, R. (2014). *The essentials of risk management* (2nd ed.). McGraw-Hill Education.

5. [Method of conduct of the course]

6. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
Foundation of Financial Risk	L1-L5: Introduction to financial risk; Market, Credit, Liquidity, Operational, Interest Rate, Business, Bankruptcy, Default, Settlement and Systemic Risk; Enterprise Risk; Risk–Return Trade-off; Asset Pricing Models (CAPM, Multifactor Models including Fama–French Models).	Class notes	a. Identify and distinguish various sources of financial risk. b. Explain the role of risk management in financial decision-making. c. Understand the relationship between risk, return and asset pricing.
Financial Institutions and Regulatory Framework	L6-12 Functions and workings of financial institutions: banks, hedge funds, mutual funds etc. Risk management practices in FIs. Basel Accord (Basel-I, II, III), Basel III capital structure (CET1, AT1 and Tier II Capital), Capital Adequacy Ratio (CAR), Capital Conservation Buffer (CCB), Countercyclical Capital Buffer (CCyB), leverage ratio, Liquidity Coverage Ratio (LCR), Net Stable Funding Ratio (NSFR), Pillar I, Pillar II and Pillar III requirements, Internal Capital Adequacy Assessment Process (ICAAP), Internal Liquidity Adequacy Assessment Process (ILAAP), stress testing and scenario analysis, market-risk reforms under the Fundamental Review of the Trading Book (FRTB), and the role of Basel regulations in strengthening the resilience and stability of the global financial system.	Ch. 2, 4, 11, 15, 16 of TB1+ Class Notes	a. Explain the role of financial institutions in the financial system. b. Understand regulatory capital and liquidity requirements under Basel III. c. Evaluate how regulatory standards enhance the resilience of financial institutions.



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Financial Risk Measurement and Analytics	L13-16 Probability distributions; Statistical concepts for risk modelling; Time-series properties of financial returns; Stochastic Processes; Geometric Brownian Motion; Itô's Lemma; Black–Scholes–Merton Model; Numerical methods and interpolation techniques; L17-27 Risk modeling – Value at risk, C-VaR, Marginal VaR, Component VaR, Options VaR, Fixed income VaR, Credit VaR, scenarios analysis and stress testing, back testing.	Class notes +Ch. 8, 9, 12 of TB1	a. Develop quantitative models for measuring financial risk. b. Apply statistical and stochastic techniques to estimate portfolio risk. c. Evaluate the effectiveness of different risk measures and risk models.
Risk Management and Hedging Strategies	L28-35 Risk management using option Greeks, implied volatility, volatility smile, volatility surface, volatility modeling, VIX modeling and application. L36-40 Risk management of fixed income securities. (Optional)	Ch. 5, 6, 7, 1, 14 of TB1 + Class notes	a. Apply derivative-based hedging techniques for managing financial risk. b. Analyse volatility dynamics and interest-rate sensitivity of financial assets. c. Assess the risk profile of financial institutions and investment portfolios using quantitative risk management techniques.

7. Evaluation Scheme:

Components	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	25%	As per time-table	Close Book/Open Book
Comprehensive Examination	40%	As per time-table	Partly Open Book
Quiz-1	15%	TBD	Close Book
Assignment	20%	TBD	Open Book

8. Minimum Criteria for a valid grade: A student should obtain 30% of the average of the top few performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.

9. Chamber consultation hours: Thursday- from 4:00PM to 5:00 PM,
Room No. 6165 F (Dept. of Economics & Finance)



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10. **Makeup Policy:** No make-up will be provided for tests unless prior approval has been obtained from the Instructor-in-Charge (I/C). Make-up for comprehensive examinations will be subject to the decision of AUGSD. The course will strictly adhere to the guidelines prescribed by AUGSD for the award of final grades.

B. Debata

Prof. Byomakesh Debata
FIN F414